Understanding Gene Behavior Across Cancer Types

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Problem / Question

For this project, I wanted to look at gene expression patterns, variability, and relationships across different cancer types. The goal was to see which genes are consistently highly expressed, which are highly variable, and which genes are central in cancer networks.

Understanding these are important because highly expressed or highly connected genes might be key players in cancer biology. Such insights could point to important pathways, potential targets for treatment and genes that respond to changes in the tumor environment.

Why This Analysis Matters

Irrespective of type, Cancer is a result of mutation in gene expressions. As a result, gene expression tells us a lot about how cells behave. Genes that are consistently active can be essential for the cell's survival or cancer growth. Genes that vary a lot might be responding to stress, the environment, or other factors, and understanding that can help us know which genes are flexible or sensitive in cancer.

Central genes in networks on the other hand, often have the ability to control important pathways, as such finding them can highlight which genes are driving major processes. Past studies (Cancer Genome Atlas Research Network et al., 2013) also found that looking at expression levels, variability, and gene networks is important to understand how cancer works.

Data Source

For this analysis, I randomly sampled a subset of genes from the Cancer Genome Atlas Pan-Cancer dataset. The data was accessed through Nyantudre (2025) on Kaggle, which is originally based on the Cancer Genome Atlas Research Network et al. (2013).

The dataset includes measurements from five cancer types: Breast Invasive Carcinoma (BRCA), Colon Adenocarcinoma (COAD), Kidney Renal Clear Cell Carcinoma (KIRC), Lung Adenocarcinoma (LUAD), and Prostate Adenocarcinoma (PRAD).

The dataset has a total of 801 samples. It originally had 20,000+genes that were narrowed down to 15 via random sampling for the easy of visualizations.

Results

Sample-Wide Gene Expression

Looking at all the samples together, some genes stood out.

- G2452, G2362, and G3427 were highly expressed on average, while G2359, G2361, and G2421 were very low or almost silent.
- Genes like G1, G2, G3, G59, G60, and G61 were moderately expressed but varied in how stable they were.
- G61 had the highest variability, meaning its expression changed a lot across samples.
- Genes like G1 and G2452 were very stable.
- G2361 had many outliers, showing occasional extreme values, while G1, G61, and G2452 were very consistent.
- The cancer type most frequent in the sample was BRCA (300 cases) while COAD (78) was the least frequent cancer.

Gene Expression by Cancer Type

An analysis of gene expression by cancer type was also conducted. Here, instead of looking at sample wide data, a filter of different types of cancer was applied to measure gene Expression with respect to each type of cancer.

The analysis showed that some patterns were consistent across all cancers.

- G2452 was the most highly expressed in BRCA, COAD, LUAD, and PRAD.
  - KIRC was a little different, with G2362 being slightly more expressed than G2452.
- G2359 is silent in all types of cancer but KIRC.
- G61 was the most variable in all cancers, responding strongly to differences in samples.

In general, cancers with higher expressions like LUAD and BRCA also had higher variability, while PRAD had low expression and low variability.

Sample-Wide Gene Correlation

Looking at how genes relate to each other across all samples, G1 and G2 were the strongest positively correlated ($r \approx 0.534$), showing that they likely work together. G3 and G3427 also moved together ($r \approx 0.421$).

Some genes also showed strong negative relationships, like G2362 and G3427 ($r \approx -0.384$) and G2 with G61 ($r \approx -0.25$).

G3 throughout the sample appeared central in the network because it had many strong positive and negative connections with other gene. Gene_2359, on the other hand, was mostly independent.

Gene Correlation by Cancer Type

In **COAD**, the strongest positive correlation was between G2452 and G61 (R = 0.369). The strongest negative correlation was between G2364 and G3 (R = -0.341). G3 was the most connected gene, while G1 was the least connected.

In **BRCA**, G1 and G2 formed the main co-expression pair, showing a strong positive correlation (R= 0.488). For this cancer, the negative links between genes included, G1 with G2452 (R =0.253). G3427 was the most central gene in the network, while G2359 was largely independent.

For **KIRC**, G59, G60, G1, and G2 formed a core co-expression module. The strongest positive correlation was between G59 and G60 (R= 0.532). G2362 showed negative correlations with this module while G59 was the most central gene in KIRC

In **LUAD**, G1 and G2 had the strongest positive correlation (R = 0.578). G2363 and G2364 also formed a tight co-expression cluster (R=0.482). For this type of cancer, G3427 drove the strongest negative correlations and often acted as a regulatory sink. It was the most central gene, while lowly expressed or highly variable genes like G2359 and G61 were the least connected.

In **PRAD**, G1 and G2 were positively correlated (R= 0.436). G1 was the most central gene here, while G3427 had limited influence on the network.

Discussion

The results of my analysis of the subset of the Cancer Genome Atlas Pan-Cancer dataset match the patterns that the original study reported. G2452 and G2362 are consistently highly expressed, and G61 is always the most variable among all cancers. Some core co-expression modules included, G1 and G2; strong antagonistic pairs, like G1 with G2452, were also seen.

Challenges

Some challenges I faced included working with only a small set of genes, which may have missed certain interactions, and handling genes with many outliers, like G2361, which can influence correlation results.

Creating the correlations in Tableau was also tricky. For the overall sample, I had to calculate the correlations in Python first, then import the results and pivot the dataset to visualize it. For the cancer-type-specific correlations, in addition to calculating them, I had to convert the dataset from wide to long format before graphing.

Coordinating the different sheets in Excel could also be challenging at times. Despite these difficulties, I was able to chart the data set and analyze it using different visualizations to observe the overall trends in expression, variability, and gene network connections.

Conclusion

Overall, this project's analysis of the Cancer Genome Atlas Pan-Cancer dataset was able to highlight some clear patterns in gene expression, variability, and network connectivity across multiple cancer types. It found certain genes, like G2452 and G2362 that are consistently highly expressed, marking them as potentially essential players in cancer biology. G61 stands out as the most variable gene across all cancers, suggesting it is highly responsive to biological or environmental differences. Co-expression patterns reveal core modules, such as G1 and G2 appear to be central in many cancers, while some genes like G2359 remain mostly independent. Strong antagonistic relationships such as G1 with G2452 also emerged consistently, showing that gene interactions can include both cooperation and repression.

Despite working with only a subset of genes and facing challenges with outliers and data formatting, the overall trends are clear and align with patterns reported in previous studies. These results reinforce that both expression levels and variability matter, and network connectivity can help highlight the most influential genes in each cancer type. This insight can guide future research on which genes might be important for cancer behavior or therapy.

Citations

Cancer Genome Atlas Research Network, Weinstein, J. N., Collisson, E. A., Mills, G. B., Shaw, K. R., Ozenberger, B. A., Ellrott, K., Shmulevich, I., Sander, C., & Stuart, J. M. (2013). The Cancer Genome Atlas Pan-Cancer analysis project. *Nature Genetics, 45*(10), 1113–1120. https://pmc.ncbi.nlm.nih.gov/articles/PMC3919969/

Nyantudre, A. (2025). *Gene Expression Cancer RNA-Seq [Data set]*. UCI Machine Learning Repository. https://www.kaggle.com/datasets/waalbannyantudre/gene-expression-cancer-rna-seq-donated-on-682016